QSA Industries ERP — Complete Workflow Guide

From setup to daily operations. This document explains how to set up the ERP, what data goes where, how every department works, and the full chain of execution for an order — from the moment a dealer places it to delivery, invoicing and payment. Screenshots are included throughout.

1. Setup

Prerequisites

- XAMPP at `C:\xampp` (MariaDB 10.11 + Apache + phpMyAdmin).
- Node.js v26+.
- Project at `D:\PROJECTS+\QSAINDUSTRIES\QSAINDUSTRIES`.

Start services

```
Start-Process -FilePath "C:\xampp\mysql\bin\mysqld.exe" -ArgumentList "--defaults-file=C:\xampp\mysql\bin\my.ini" -WindowStyle Hidden
Start-Process -FilePath "C:\xampp\apache\bin\httpd.exe" -WindowStyle Hidden
```

Start app

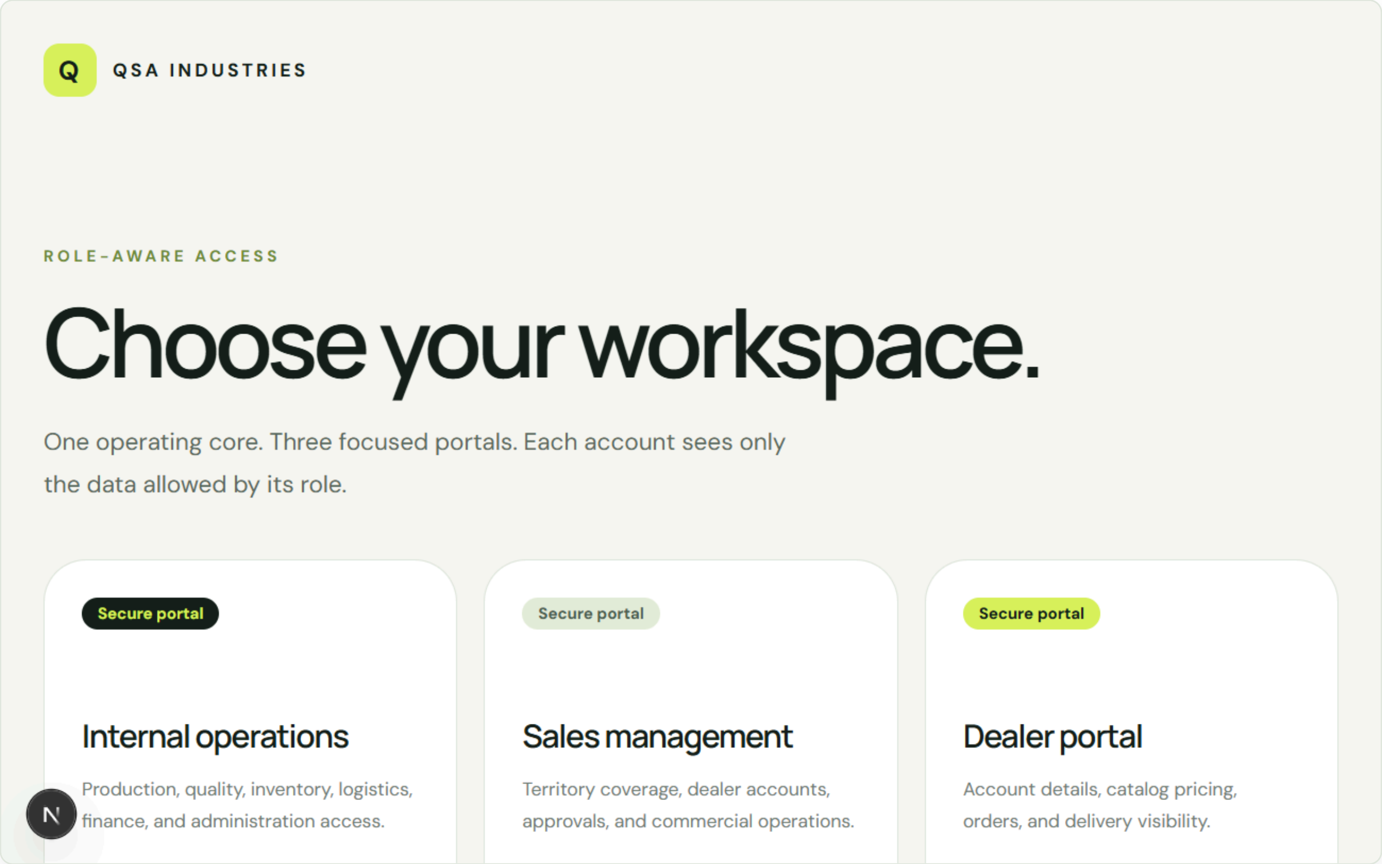
```
cd D:\PROJECTS+\QSAINDUSTRIES\QSAINDUSTRIES\web
npm run dev
```

App: `http://localhost:3000` | Login: `http://localhost:3000/login` | phpMyAdmin: `http://localhost/phpmyadmin`

Credentials

Account	Email	Password
Admin	admin@qsa-demo.in	Qsa@Demo2026
Finance	finance@qsa-demo.in	Qsa@Demo2026
Sales Director	nadia.kapoor@qsa-demo.in	Qsa@Demo2026
Production	operations.production@qsa-demo.in	Qsa@Demo2026
Inventory	operations.inventory@qsa-demo.in	Qsa@Demo2026
QC	qc@qsa-demo.in	Qsa@Demo2026
Logistics	logistics@qsa-demo.in	Qsa@Demo2026
Dealer	dealer@qsa-demo.in	Qsa@Demo2026

MySQL: host `localhost:3306`, database `qsa_erp`, user `qsa_user` / `Qsa@Demo2026`.



Q QSA INDUSTRIES

ROLE-AWARE ACCESS

Choose your workspace.

One operating core. Three focused portals. Each account sees only the data allowed by its role.

Secure portal

Internal operations

Production, quality, inventory, logistics, finance, and administration access.

Secure portal

Sales management

Territory coverage, dealer accounts, approvals, and commercial operations.

Secure portal

Dealer portal

Account details, catalog pricing, orders, and delivery visibility.

Login screen — choose Internal / Sales / Dealer portal.

2. Architecture (local mode)

- `web/src/lib/db.ts` — mysql2 pool (env MYSQL_HOST/PORT/USER/PASSWORD/DATABASE).
- `web/src/lib/auth.ts` — cookie session `qsa_session` (HMAC-signed), `getProfile()` reads `users` table.
- `web/src/app/api/auth/login|logout` — bcrypt verify + cookie set/clear.
- `web/src/lib/supabase/*` — shims returning empty data for unmigrated reads (non-crashing).
- `web/src/proxy.ts` — CSRF Origin check + global rate limit (10 req/min/IP).
- Each module has dedicated routes under `web/src/app/api/...`.

3. User roles

Role	Portal	Scope
admin	Internal	Full access; superuser passes every guard.
sales_director / sales_manager	Sales	Approve/escalate orders, view reports.
sales_rep	Sales	Raise proxy orders for mapped dealers, edit pending orders.
finance	Internal	Invoices, payments, expenses, credit, Tally export.
production_manager	Internal	Production runs, QC.
inventory_manager	Internal	Stock, adjustments, purchase orders.
qc_staff	Internal	QC verdicts.
logistics	Internal	Dispatch & delivery.
dealer	Dealer	Catalog, own orders, own invoices.

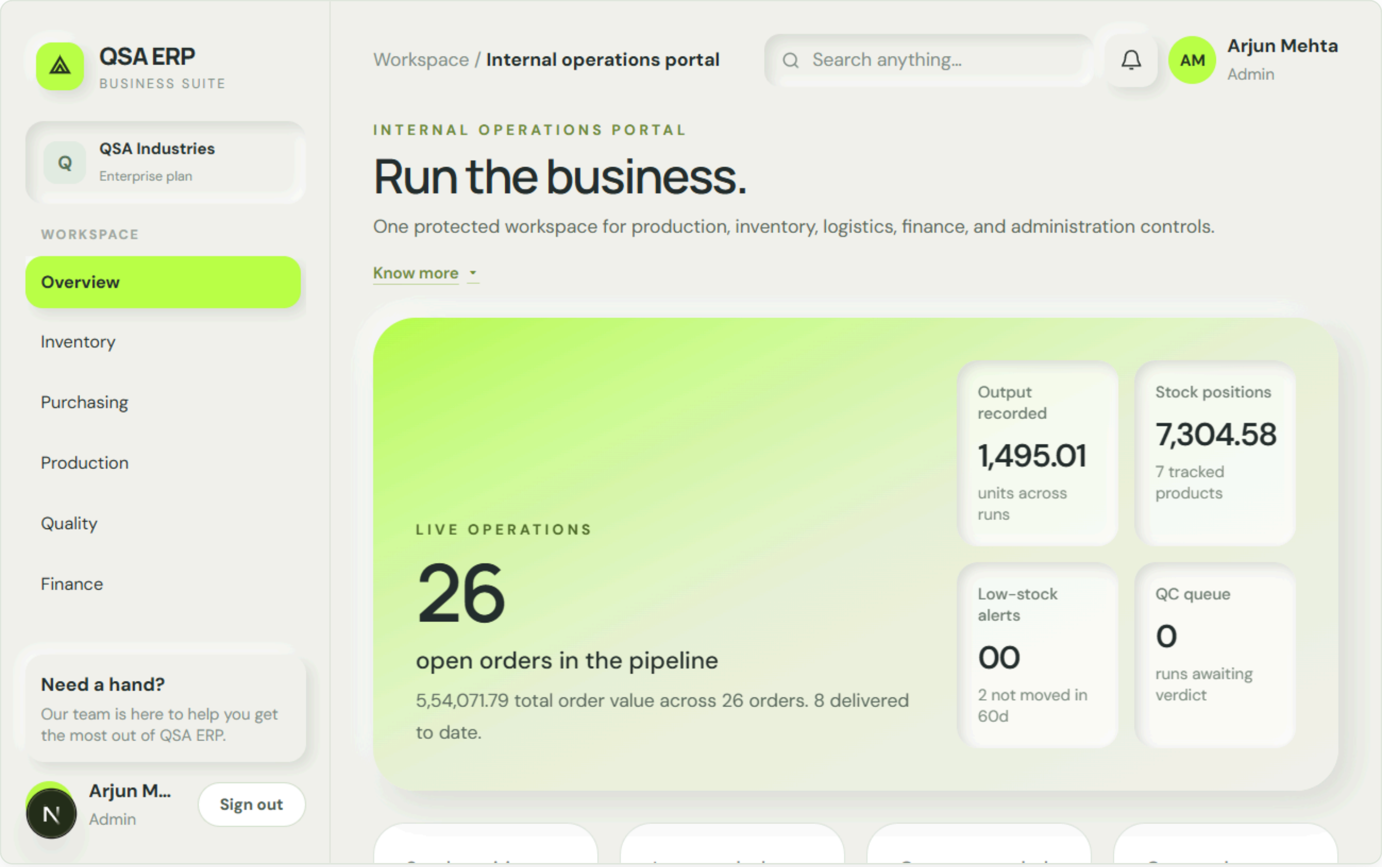
4. Database (core tables)

`users, dealers, products, price_lists, orders, order_items, invoices, payments, stock_levels, stock_movements, production_runs, production_batches, purchase_orders, purchase_order_lines, finance_settings, expense_heads, expense_ledger, designations, org_positions, dealer_onboardings, tally_exports.`

`users.password_hash` holds bcrypt hashes; `users.reports_to_id` drives the org chart.

5. Navigation & UI

Sidebar order: Overview, Inventory, Purchasing, Production, Quality, Finance, Dispatch queue, Delivery board, Organization, Dealer onboarding, Sales orders, Reporting, Settings.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LIVE OPERATIONS
26
Output recorded
1,495.01
units across runs
Stock positions
7,304.58
7 tracked products
Low-stock alerts
00
2 not moved in 60d
QC queue
0
runs awaiting verdict

Run the business.

One protected workspace for production, inventory, logistics, finance, and administration controls.

open orders in the pipeline

5,54,071.79 total order value across 26 orders. 8 delivered to date.

Need a hand?

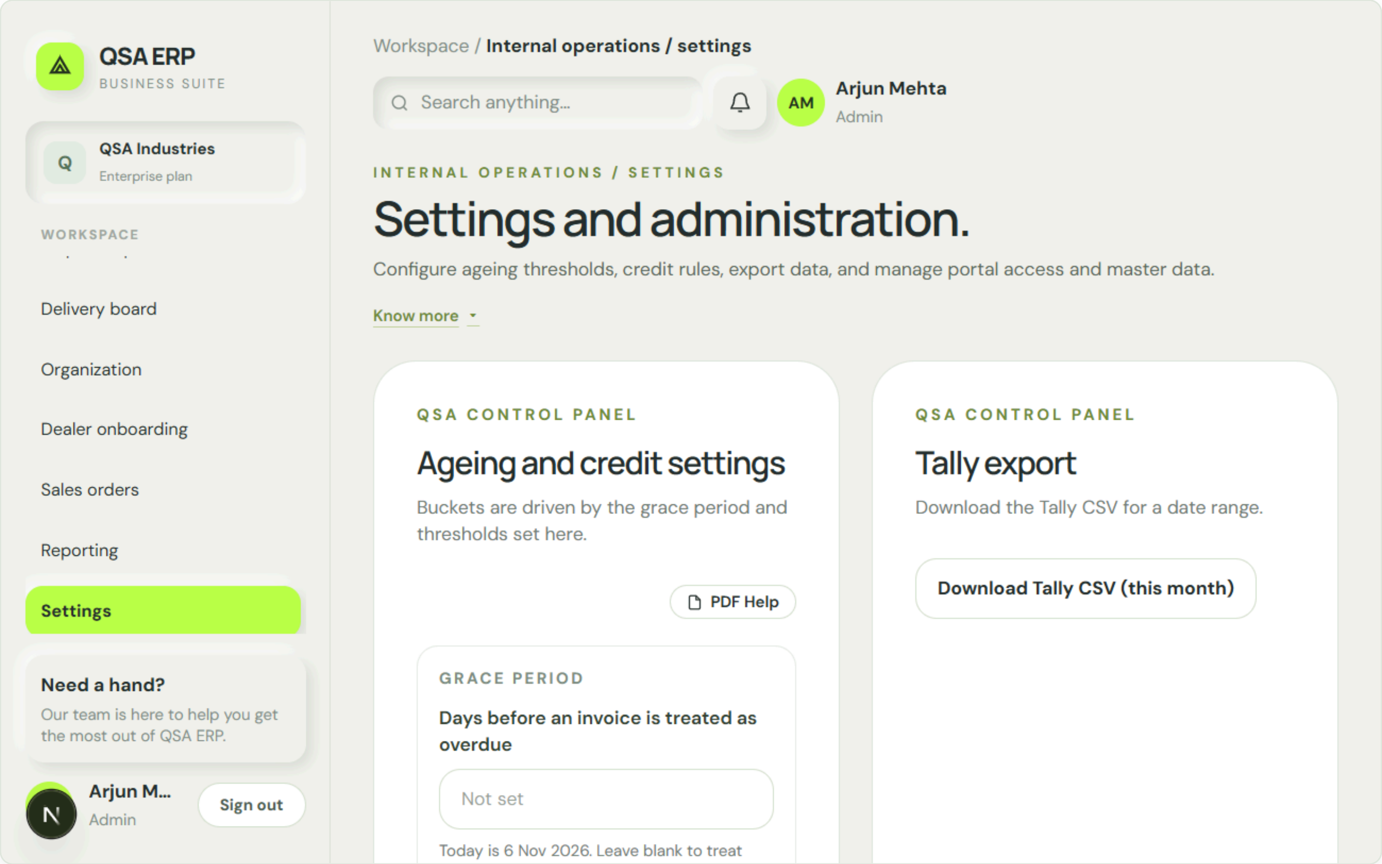
Our team is here to help you get the most out of QSA ERP.

Internal overview — KPI hero, charts, order board.

6. Department workflows

6.1 Admin

Products, dealers, users, designations, price lists, settings, dealer onboarding, org chart. Admin is the only role that can create master data.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Download Tally CSV (this month)

Workspace / Internal operations / settings

INTERNAL OPERATIONS / SETTINGS

Settings and administration.

Configure ageing thresholds, credit rules, export data, and manage portal access and master data.

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QSA CONTROL PANEL

Ageing and credit settings

Buckets are driven by the grace period and thresholds set here.

GRACE PERIOD

Days before an invoice is treated as overdue

Not set

Today is 6 Nov 2026. Leave blank to treat

QSA CONTROL PANEL

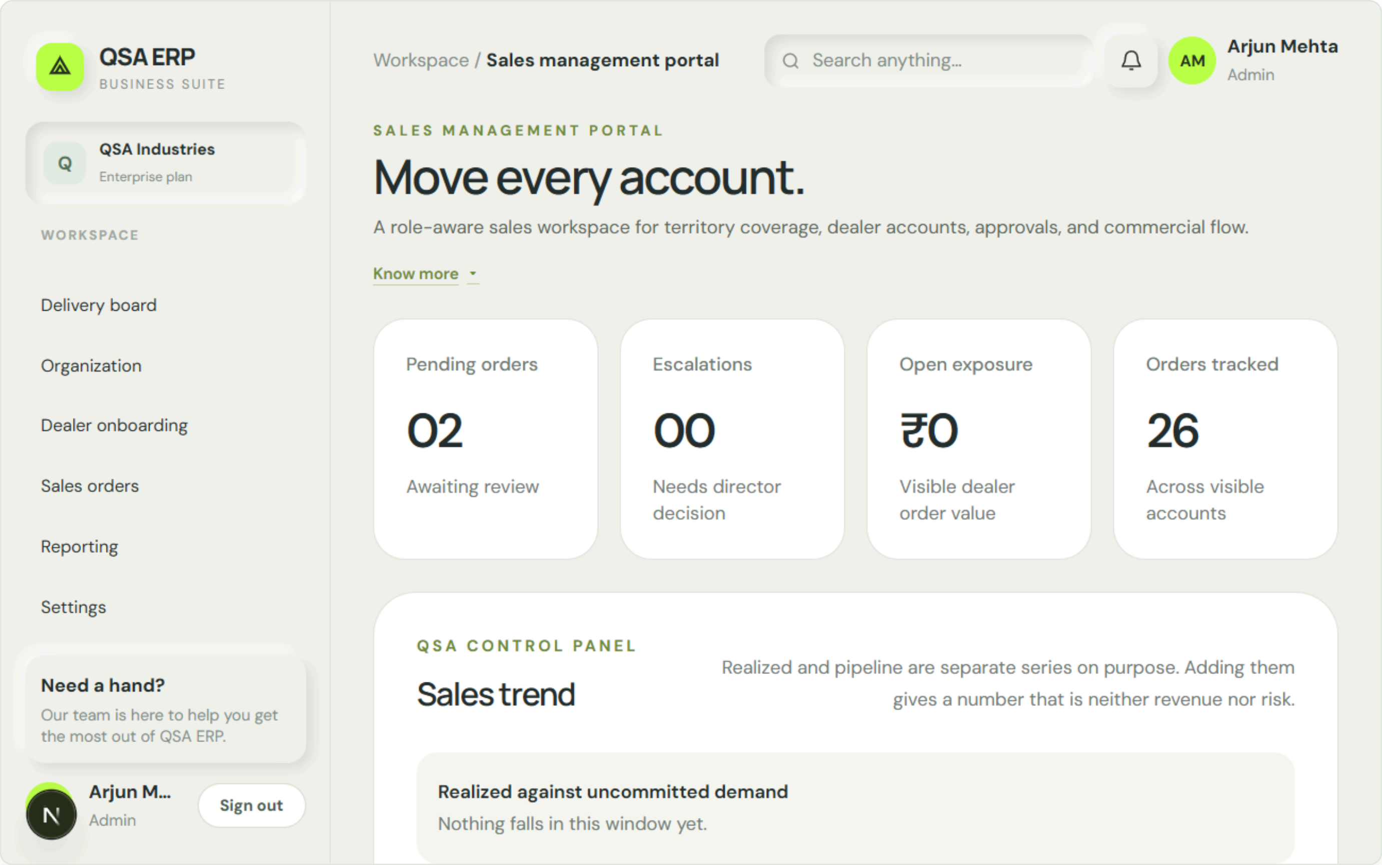
Tally export

Download the Tally CSV for a date range.

Settings page — ageing/credit, Tally export, portal quick links.

6.2 Sales

Sales reps manage dealers mapped to them, raise proxy orders, approve/escalate orders. Sales managers/directors approve escalated orders.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Workspace / Sales management portal

SALES MANAGEMENT PORTAL

Move every account.

A role-aware sales workspace for territory coverage, dealer accounts, approvals, and commercial flow.

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Pending orders	Escalations	Open exposure	Orders tracked
02	00	₹0	26
Awaiting review	Needs director decision	Visible dealer order value	Across visible accounts

QSA CONTROL PANEL

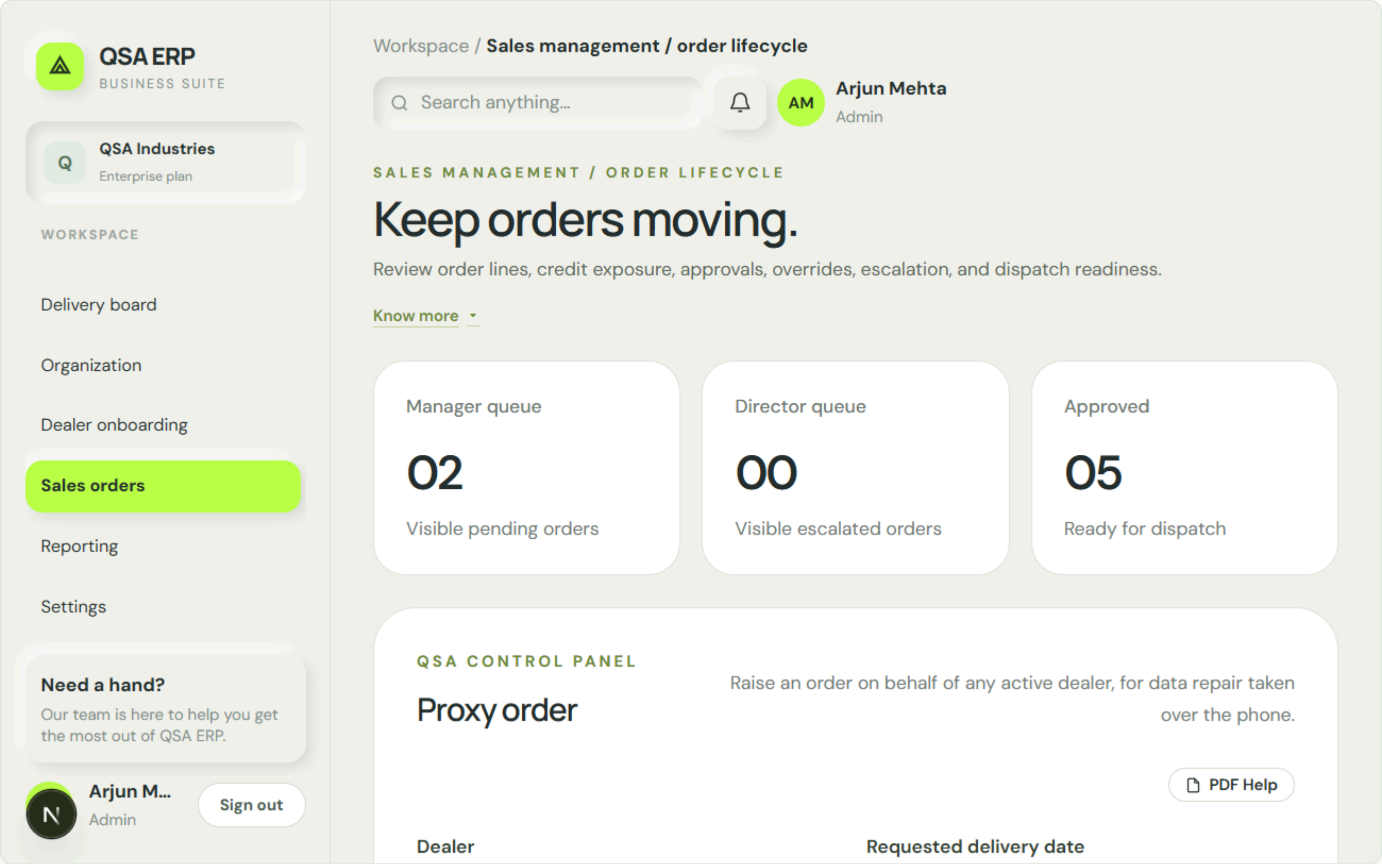
Sales trend

Realized and pipeline are separate series on purpose. Adding them gives a number that is neither revenue nor risk.

Realized against uncommitted demand

Nothing falls in this window yet.

Sales dashboard.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SALES MANAGEMENT / ORDER LIFECYCLE

Keep orders moving.

Review order lines, credit exposure, approvals, overrides, escalation, and dispatch readiness.

Manager queue
02
Visible pending orders

Director queue
00
Visible escalated orders

Approved
05
Ready for dispatch

QSA CONTROL PANEL

Proxy order

Raise an order on behalf of any active dealer, for data repair taken over the phone.

Dealer

Requested delivery date

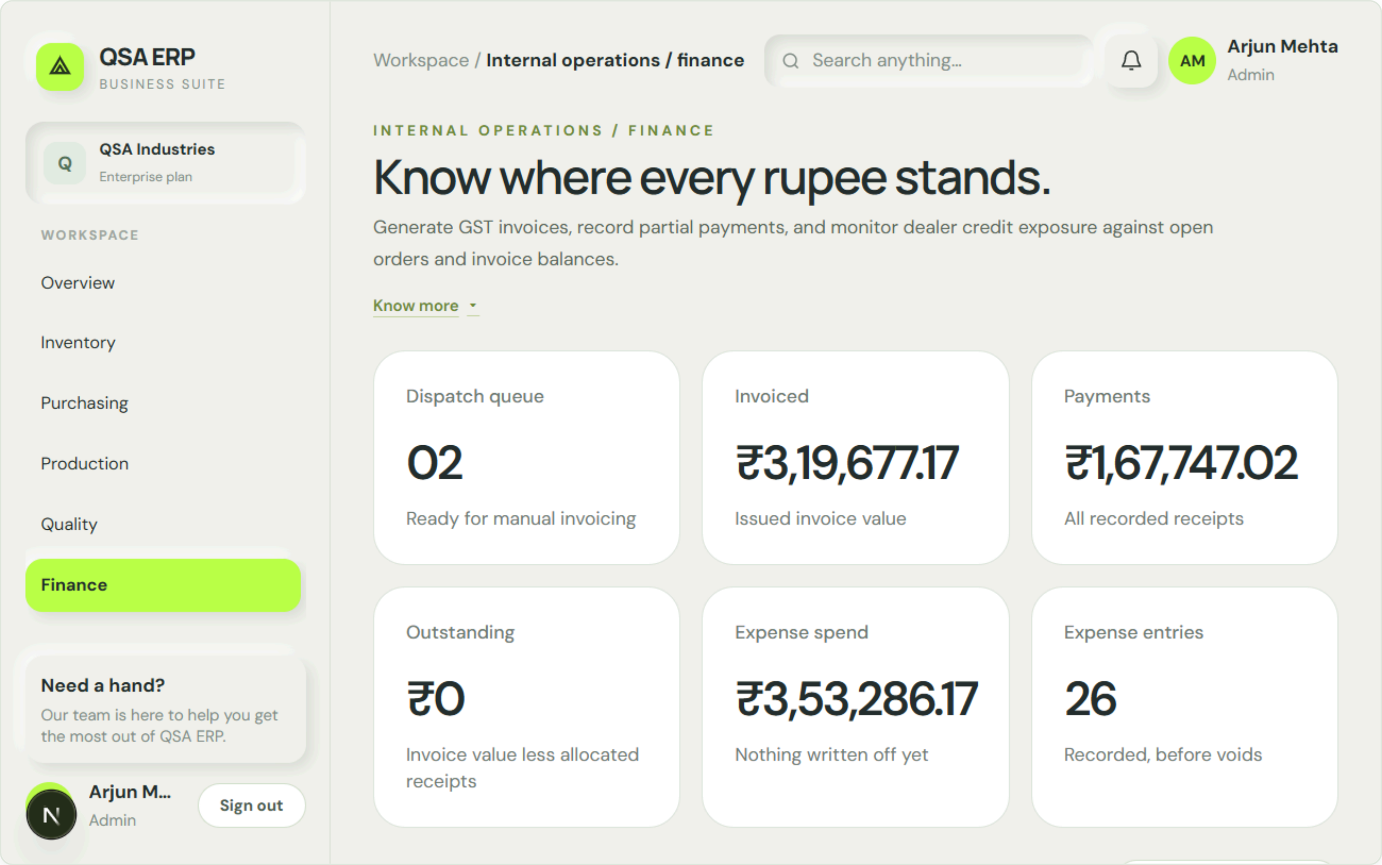
Need a hand?

Our team is here to help you get the most out of QSA ERP.

Sales orders — queue, proxy order form, edit.

6.3 Finance

Generate invoices, record payments, manage dealer credit, record expenses, export to Tally, configure ageing/credit settings.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INTERNAL OPERATIONS / FINANCE

Know where every rupee stands.

Generate GST invoices, record partial payments, and monitor dealer credit exposure against open orders and invoice balances.

Dispatch queue
02
Ready for manual invoicing

Invoiced
₹3,19,677.17
Issued invoice value

Payments
₹1,67,747.02
All recorded receipts

Outstanding
₹0
Invoice value less allocated receipts

Expense spend
₹3,53,286.17
Nothing written off yet

Expense entries
26
Recorded, before voids

Need a hand?

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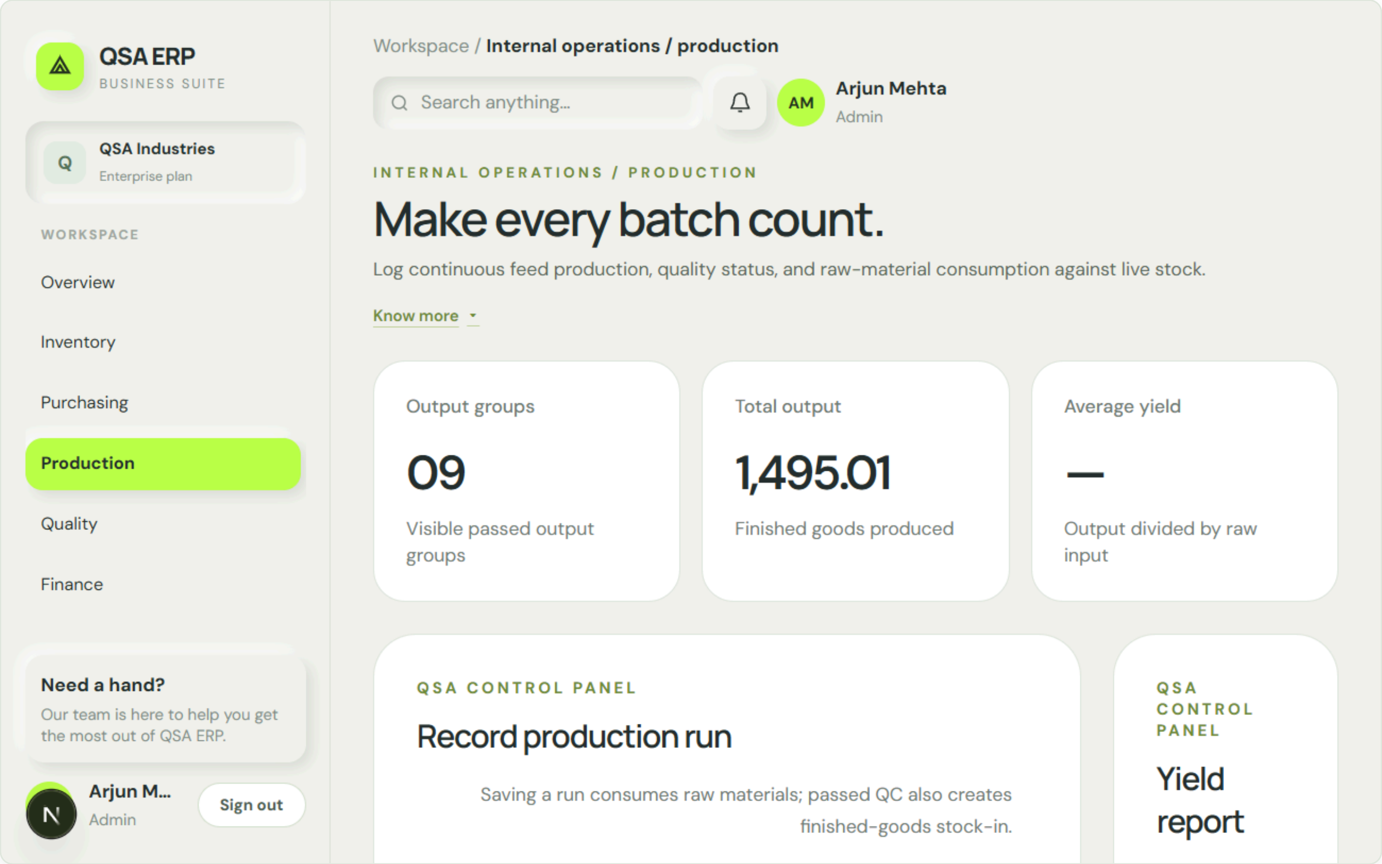
Finance page — invoices, payments, expenses, exposure.

6.4 Production

Create production runs (finished goods from raw materials), QC verdicts, batch tracking with expiry.



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INTERNAL OPERATIONS / PRODUCTION

Make every batch count.

Log continuous feed production, quality status, and raw-material consumption against live stock.

Output groups: 09 — Visible passed output groups

Total output: 1,495.01 — Finished goods produced

Average yield: — — Output divided by raw input

QSA CONTROL PANEL

Record production run

Saving a run consumes raw materials; passed QC also creates finished-goods stock-in.

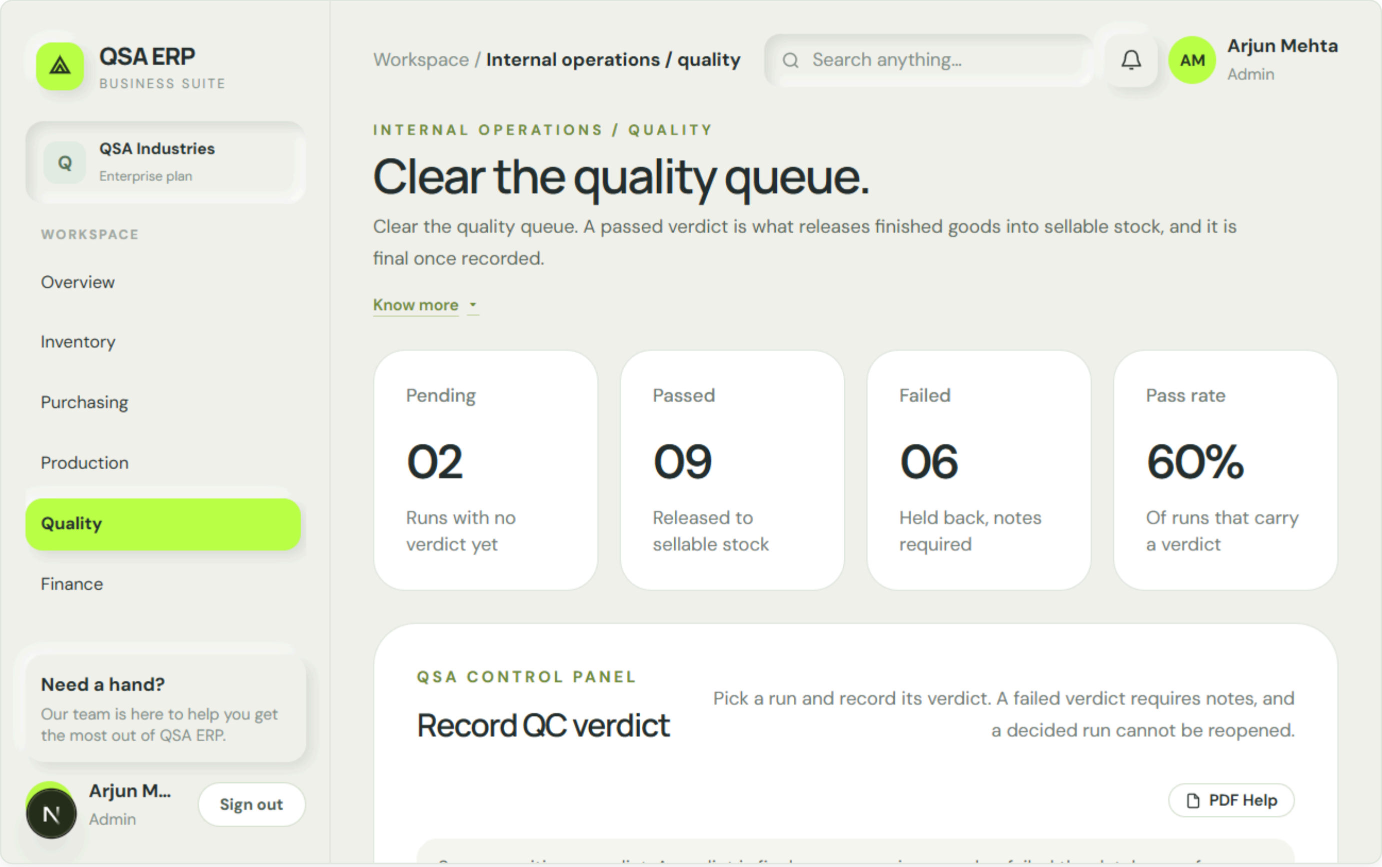
QSA CONTROL PANEL

Yield report

Production page.

6.5 Quality

QC staff record pass/hold/fail on pending runs.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INTERNAL OPERATIONS / QUALITY

Clear the quality queue.

Clear the quality queue. A passed verdict is what releases finished goods into sellable stock, and it is final once recorded.

Pending	Passed	Failed	Pass rate
02	09	06	60%
Runs with no verdict yet	Released to sellable stock	Held back, notes required	Of runs that carry a verdict

QSA CONTROL PANEL

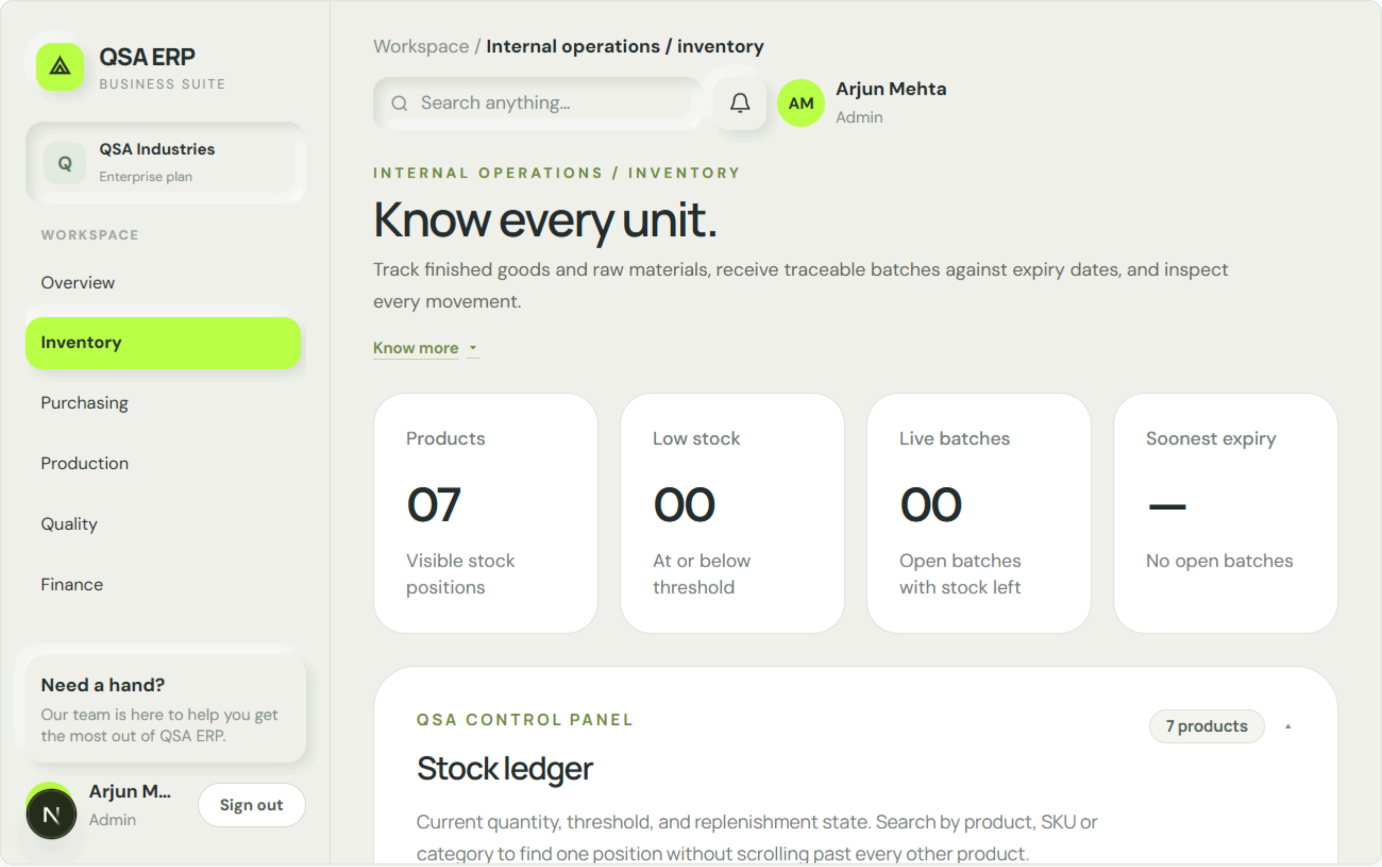
Record QC verdict

Pick a run and record its verdict. A failed verdict requires notes, and a decided run cannot be reopened.

Quality page.

6.6 Inventory

Stock levels, adjustments, batch receive, low-stock & dead-stock signals.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7 products

INTERNAL OPERATIONS / INVENTORY

Know every unit.

Track finished goods and raw materials, receive traceable batches against expiry dates, and inspect every movement.

Products	Low stock	Live batches	Soonest expiry
07	00	00	—
Visible stock positions	At or below threshold	Open batches with stock left	No open batches

QSA CONTROL PANEL

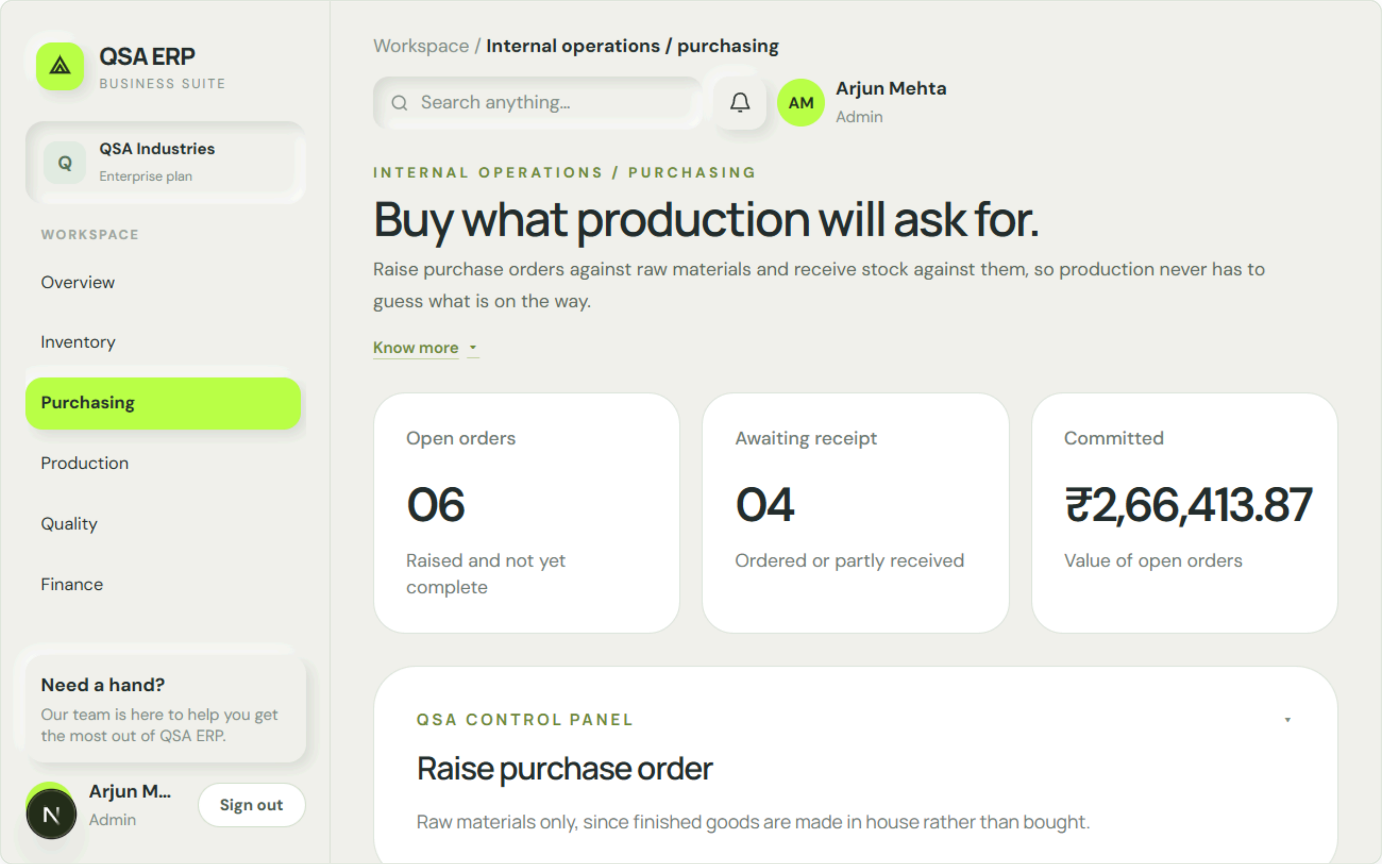
Stock ledger

Current quantity, threshold, and replenishment state. Search by product, SKU or category to find one position without scrolling past every other product.

Inventory page.

6.7 Purchasing

Purchase orders for raw materials, receiving, supplier tracking.



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Workspace / **Internal operations / purchasing**

INTERNAL OPERATIONS / PURCHASING

Buy what production will ask for.

Raise purchase orders against raw materials and receive stock against them, so production never has to guess what is on the way.

Open orders	Awaiting receipt	Committed
06	04	₹2,66,413.87
Raised and not yet complete	Ordered or partly received	Value of open orders

QSA CONTROL PANEL

Raise purchase order

Raw materials only, since finished goods are made in house rather than bought.

Need a hand?

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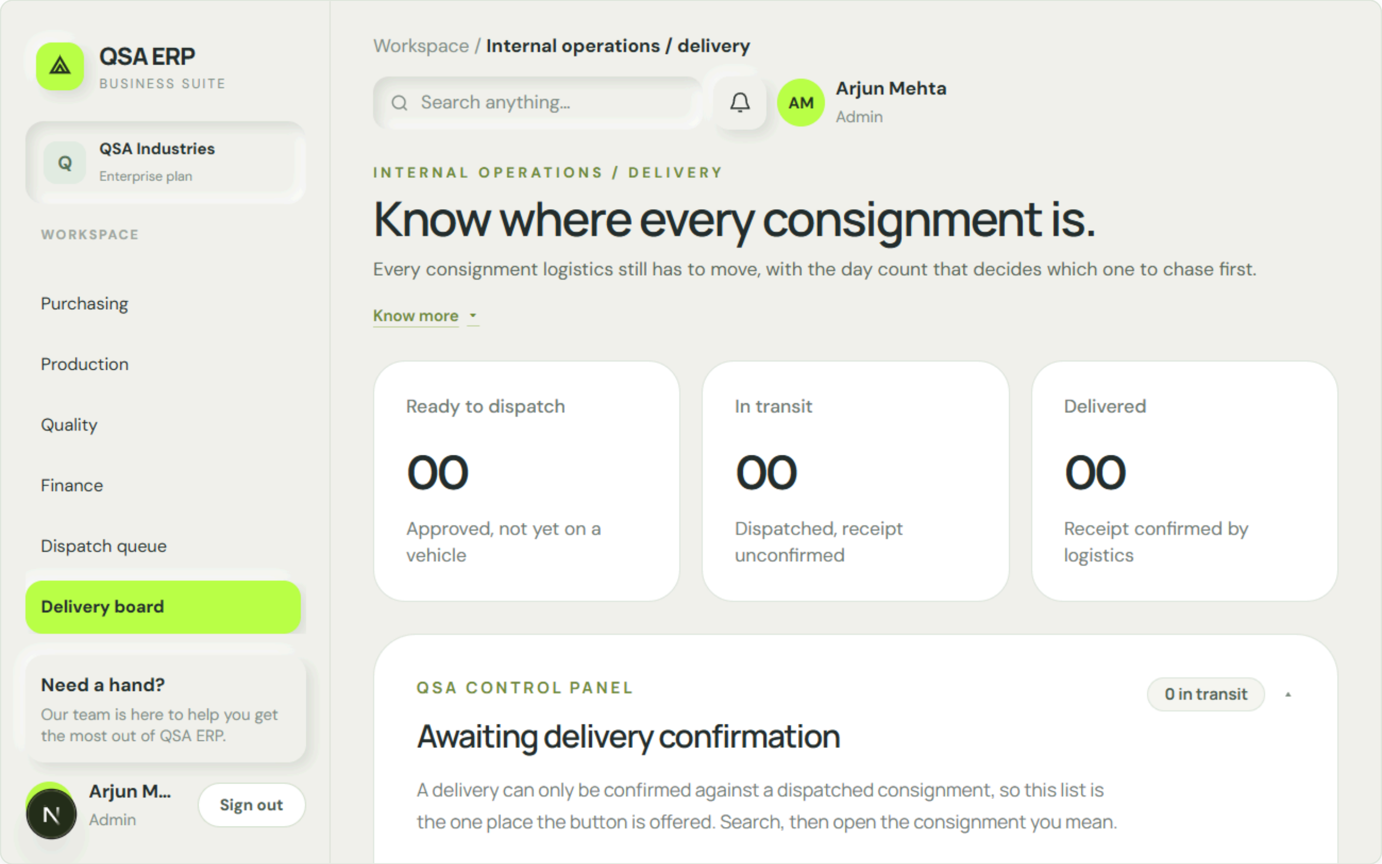
Purchasing page.

6.8 Logistics

Dispatch queue and delivery board: mark dispatched, record tracking, confirm delivery.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INTERNAL OPERATIONS / DELIVERY

Know where every consignment is.

Every consignment logistics still has to move, with the day count that decides which one to chase first.

Know more

Ready to dispatch	In transit	Delivered
00	00	00
Approved, not yet on a vehicle	Dispatched, receipt unconfirmed	Receipt confirmed by logistics

QSA CONTROL PANEL

0 in transit

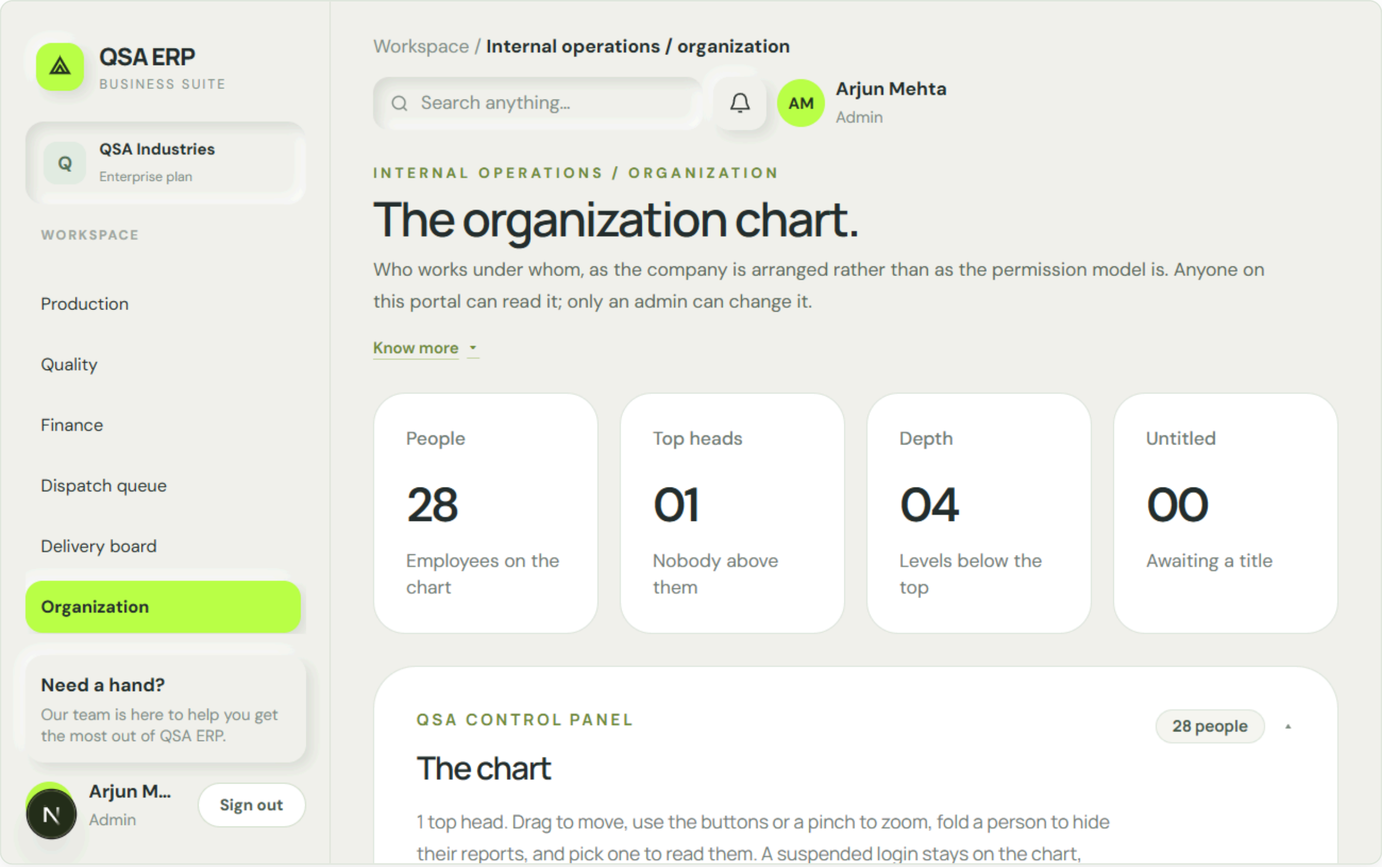
Awaiting delivery confirmation

A delivery can only be confirmed against a dispatched consignment, so this list is the one place the button is offered. Search, then open the consignment you mean.

Delivery board.

6.9 Organization

Org chart and people builder: positions, reporting lines, designations.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INTERNAL OPERATIONS / ORGANIZATION

The organization chart.

Who works under whom, as the company is arranged rather than as the permission model is. Anyone on this portal can read it; only an admin can change it.

Know more

People	Top heads	Depth	Untitled
28	01	04	00
Employees on the chart	Nobody above them	Levels below the top	Awaiting a title

QSA CONTROL PANEL

28 people

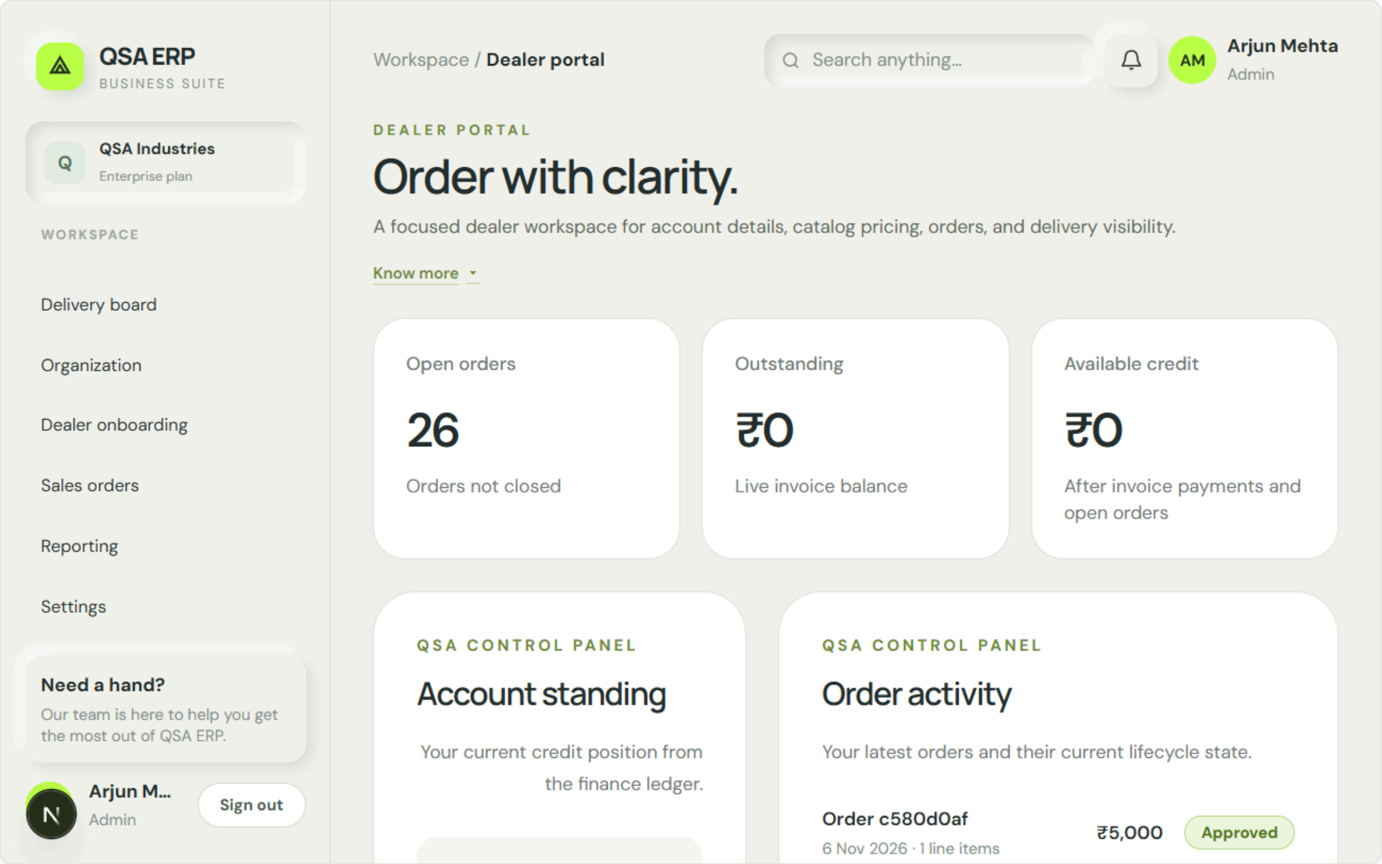
The chart

1 top head. Drag to move, use the buttons or a pinch to zoom, fold a person to hide their reports, and pick one to read them. A suspended login stays on the chart,

Organization chart.

6.10 Dealer portal

Dealers browse catalog, place orders, view invoices and delivery status.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DEALER PORTAL

Order with clarity.

A focused dealer workspace for account details, catalog pricing, orders, and delivery visibility.

Open orders
26
Orders not closed

Outstanding
₹0
Live invoice balance

Available credit
₹0
After invoice payments and open orders

Need a hand?
Our team is here to help you get the most out of QSA ERP.

QSA CONTROL PANEL
Account standing
Your current credit position from the finance ledger.

QSA CONTROL PANEL
Order activity
Your latest orders and their current lifecycle state.

Order c580d0af
6 Nov 2026 · 1 line items
₹5,000
Approved

Dealer portal.

7. Order lifecycle — chain of execution

An order flows through these states. Each transition is a deliberate action by a specific role.

State	Who acts	What happens
pending	Dealer / Sales rep (proxy)	Order created with lines, stock checked, price resolved. Status pending.
approved	Sales manager / admin	Approve if within credit limit. Moves to dispatch queue.
pending_director_approval	Sales rep escalates	Order exceeds approval limit; escalated to director.
director_approved / rejected	Sales director	Director approves or rejects escalated order.
dispatched	Logistics	Mark dispatched with date and tracking note.
invoiced	Finance	Generate invoice (taxable + 5% GST). Order marked invoiced.
delivered	Logistics	Confirm delivery.
cancelled	Admin / Sales	Cancel an order.

Path A — Happy path (dealer places order)

1. Dealer logs in → catalog → selects products & quantities → places order (status **pending**).
2. Sales rep/manager opens Sales → Orders → reviews lines & credit exposure.
3. If within approval limit: **Approve** → status **approved** → appears in Dispatch queue.
4. Logistics marks **Dispatched** (date + tracking note).
5. Finance generates **Invoice** → status **invoiced** (taxable + 5% GST).
6. Logistics confirms **Delivered**.
7. Finance records **Payment** (partial or full) against the invoice.

Path B — Proxy order (sales rep on behalf of dealer)

1. Sales rep/admin opens Sales → Orders → "Proxy order" panel.
2. Selects dealer (must be mapped to the rep; admin can pick any active dealer) and product lines.
3. Submits → order created as **pending** with stock & price validated.
4. Continues from step 2 of Path A.

Path C — Manual sales order

1. Sales/Admin opens Sales → Orders → "Manual sales order" panel.
2. Picks an existing dealer, status, delivery date, product lines with custom prices.
3. Submits → order created tagged to the dealer (status as chosen).

Path D — Credit issue (order stuck)

1. Order exceeds the dealer's credit limit or approval threshold.
2. Sales rep clicks **Escalate** → status **pending_director_approval**.
3. Sales director reviews credit exposure and either **Approves** (→ approved) or **Rejects**.
4. Alternatively, admin sets a credit override on the dealer, then approves.
5. If credit was refused at onboarding, the dealer cannot trade until credit is approved.

Path E — Manual invoice (no order)

1. Finance opens Finance → "Manual invoice" panel.
2. Selects dealer, enters taxable/GST/total amounts (optional invoice number & date).
3. Invoice created tagged to the dealer (order_id is null) and appears in the register & Tally export.

Stuck / edge cases

- **No sales rep mapped** → dealer cannot be activated during onboarding; orders blocked.

- **Inactive dealer** → cannot place orders.
- **Insufficient stock** → order creation refused.
- **Low stock / dead stock** → flagged on overview, not blocking.
- **QC fail/hold** → production run produces nothing; not counted in yield.
- **Voided expense** → stays in register struck through; net totals exclude it.

8. Data entry — where to put what

Each data-entry screen has a **PDF Help** link (top-right of the form) opening a detailed guide in `/guides/`. Summary of where data is entered:

Data	Where	Role
Products	Admin → Product catalogue / Create a product	admin
Dealers	Admin → Dealer master / Create a dealer (or Dealer onboarding)	admin
Users / staff logins	Internal → Users	admin
Price lists	Admin → Price list / Create a price	admin
Designations	Admin → Designations manager	admin
Org chart / people	Internal → Organization	admin
Finance settings	Internal → Settings	admin/finance
Expense heads	Finance → Expense heads	admin/finance
Expenses	Finance → Record an expense	admin/finance
Sales orders (dealer)	Dealer portal → Orders	dealer
Sales orders (proxy)	Sales → Orders → Proxy order	sales_rep/admin
Sales orders (manual)	Sales → Orders → Manual sales order	sales_rep/admin
Order approval/escalation/dispatch	Sales → Orders (open an order)	sales_manager/director/logistics
Invoices	Finance → Invoice queue → Generate invoice; or Manual invoice	finance/admin
Payments	Finance → Invoice register / Dealer account payments	finance/admin
Dealer credit	Finance → Dealer credit and region	admin/finance
Production runs	Internal → Production → create run	production_manager/admin
QC verdicts	Internal → Quality	qc_staff/production_manager
Stock adjustments	Internal → Inventory → stock adjustment	inventory/production/admin
Batch receive	Internal → Inventory → batch receive	inventory/admin
Purchase orders	Internal → Purchasing → create PO	inventory/production/admin
PO receive	Purchasing → receive against PO	inventory/admin
Delivery confirm	Internal → Delivery board	logistics/admin
Dealer onboarding	Internal → Dealer onboarding	admin
Tally export	Finance → Download Tally CSV / Settings	finance/admin

9. Troubleshooting

- **MySQL not running:** start mysqld via XAMPP; app needs DB before `npm run dev`.
- **"Objects are not valid as a React child (Date)":** ensure `dateStrings: true` in `db.ts`.
- **Access denied for qsa_user:** recreate via `mysql -u root < user2.sql`.
- **Empty analytics screens:** views not yet ported to MySQL (remaining gap).
- **403 Forbidden:** role not allowed for that route.
- **429 Too Many Requests:** rate limit (10 req/min/IP) exceeded.

QSA Industries ERP — Complete Workflow Guide. Generated for local MySQL build.